

Sex discrimination case study (1974)

MATH 213

Group 1

We consider a study investigating sex discrimination in the 1970s, which is set in the context of personnel decisions within a bank. The research question we hope to answer is, “Are females discriminated against in promotion decisions made by their supervisors who are male?” (Rosen and Jerdee 1974)

The participants in this study were 48 male bank supervisors attending a management institute at the University of North Carolina in 1972. They were asked to assume the role of the personnel director of a bank and were each given one personnel file to judge whether the person should be promoted to a branch manager position. The files given to the participants were identical, except that half of them indicated the candidate was male and the other half indicated the candidate was female. These files were randomly assigned to the subjects. For each supervisor both the sex associated with the assigned file and the promotion decision were recorded.

1. Is this an observational study or an experiment?
2. How does the type of study impact what can be inferred from the results?
3. Who or what are the cases?
4. What is the explanatory variable? What is the response variable? Also say the type of each variable.

Explanatory:

Response:

5. In the Posit Cloud file, make a contingency (or two-way) table to summarize the results of the study. Remember to label the columns with the explanatory variable. Put counts (rather than proportions) as the values in your table. Paste your table here.

6. Also make an appropriate graph that summarizes the results of this study. Paste your graph here.

7. What proportion of supervisors made the decision to promote, overall? (*Write your answer as a decimal. Show work.*)
8. What proportion of supervisors in the “male file” group decided to promote? (*Write your answer as a decimal. Show work.*)
9. What proportion of supervisors in the “female file” group decided to promote? (*Write your answer as a decimal. Show work.*)
10. What is the difference in the proportion of supervisors who promoted between each group? Calculate this as “proportion of supervisors in male group who promoted” - “proportion of supervisors in female group who promoted”. (*Write your answer as a decimal. Show work.*)
11. What are your initial thoughts on these data? In particular, does it look as if individuals who identify as female are unfairly discriminated against? We will discuss this as a class.

Group 2

We consider a study investigating sex discrimination in the 1970s, which is set in the context of personnel decisions within a bank. The research question we hope to answer is, “Are females discriminated against in promotion decisions made by their supervisors who are male?” (Rosen and Jerdee 1974)

The participants in this study were 48 male bank supervisors attending a management institute at the University of North Carolina in 1972. They were asked to assume the role of the personnel director of a bank and were each given one personnel file to judge whether the person should be promoted to a branch manager position. The files given to the participants were identical, except that half of them indicated the candidate was male and the other half indicated the candidate was female. These files were randomly assigned to the subjects. For each supervisor both the sex associated with the assigned file and the promotion decision were recorded.

1. Is this an observational study or an experiment?
Experiment
2. How does the type of study impact what can be inferred from the results?
3. Who or what are the cases?
The 48 bank supervisors
4. What is the explanatory variable? What is the response variable? Also say the type of each variable.

Explanatory:

Sex of the candidate

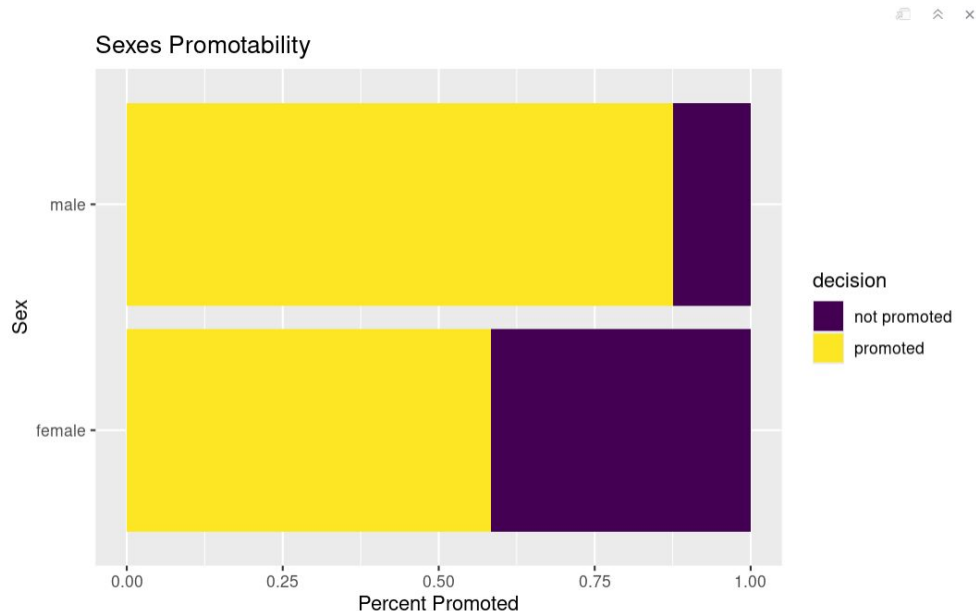
Response:

Candidate promotion

5. In the Posit Cloud file, make a contingency (or two-way) table to summarize the results of the study. Remember to label the columns with the explanatory variable. Put counts (rather than proportions) as the values in your table. Paste your table here.

decision <chr>	female <int>	male <int>
not promoted	10	3
promoted	14	21

6. Also make an appropriate graph that summarizes the results of this study. Paste your graph here.



7. What proportion of supervisors made the decision to promote, overall? (*Write your answer as a decimal. Show work.*)
8. What proportion of supervisors in the “male file” group decided to promote? (*Write your answer as a decimal. Show work.*)
9. What proportion of supervisors in the “female file” group decided to promote? (*Write your answer as a decimal. Show work.*)
10. What is the difference in the proportion of supervisors who promoted between each group? Calculate this as “proportion of supervisors in male group who promoted” - “proportion of supervisors in female group who promoted”. (*Write your answer as a decimal. Show work.*)
11. What are your initial thoughts on these data? In particular, does it look as if individuals who identify as female are unfairly discriminated against? We will discuss this as a class.

Group 3

We consider a study investigating sex discrimination in the 1970s, which is set in the context of personnel decisions within a bank. The research question we hope to answer is, “Are females discriminated against in promotion decisions made by their supervisors who are male?” (Rosen and Jerdee 1974)

The participants in this study were 48 male bank supervisors attending a management institute at the University of North Carolina in 1972. They were asked to assume the role of the personnel director of a bank and were each given one personnel file to judge whether the person should be promoted to a branch manager position. The files given to the participants were identical, except that half of them indicated the candidate was male and the other half indicated the candidate was female. These files were randomly assigned to the subjects. For each supervisor both the sex associated with the assigned file and the promotion decision were recorded.

1. Is this an observational study or an experiment?

Experiment.

2. How does the type of study impact what can be inferred from the results?

An experiment reduces confounding variables and as a result allows causality to be inferred

3. Who or what are the cases?

The cases are the 48 male supervisors.

4. What is the explanatory variable? What is the response variable? Also say the type of each variable.

Explanatory:

The sex of the applicant

Response:

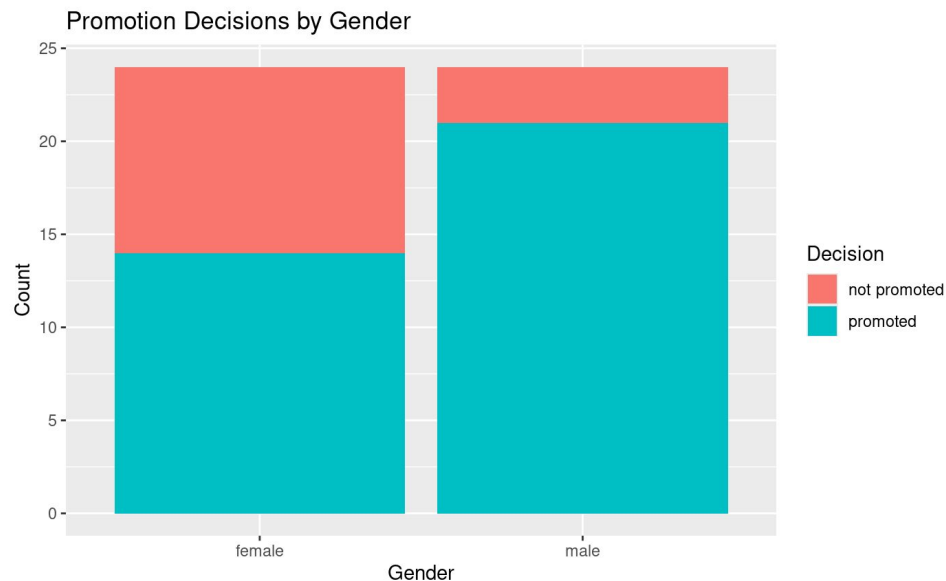
Whether or not the applicant got promoted or not

5. In the Posit Cloud file, make a contingency (or two-way) table to summarize the results of the study. Remember to label the columns with the explanatory variable. Put counts (rather than proportions) as the values in your table. Paste your table here.

A tibble: 2 × 3

decision <chr>	female <int>	male <int>
not promoted	10	3
promoted	14	21

6. Also make an appropriate graph that summarizes the results of this study. Paste your graph here.



7. What proportion of supervisors made the decision to promote, overall? *(Write your answer as a decimal. Show work.)*

$$35/48 = .7292$$

8. What proportion of supervisors in the “male file” group decided to promote? *(Write your answer as a decimal. Show work.)*

$$21/24 = .875$$

9. What proportion of supervisors in the “female file” group decided to promote? *(Write your answer as a decimal. Show work.)*

$$14/24 = .5833$$

10. What is the difference in the proportion of supervisors who promoted between each group? Calculate this as “proportion of supervisors in male group who promoted” - “proportion of supervisors in female group who promoted”. *(Write your answer as a decimal. Show work.)*

$$.875 - .5833 = .2917$$

11. What are your initial thoughts on these data? In particular, does it look as if individuals who identify as female are unfairly discriminated against? We will discuss this as a class.

It does appear that people who identified as female in the 1970s were discriminated against when it comes to promotions at banks, evident by the 30% difference in proportional promotion rates.

Group 4

We consider a study investigating sex discrimination in the 1970s, which is set in the context of personnel decisions within a bank. The research question we hope to answer is, “Are females discriminated against in promotion decisions made by their supervisors who are male?” (Rosen and Jerdee 1974)

The participants in this study were 48 male bank supervisors attending a management institute at the University of North Carolina in 1972. They were asked to assume the role of the personnel director of a bank and were each given one personnel file to judge whether the person should be promoted to a branch manager position. The files given to the participants were identical, except that half of them indicated the candidate was male and the other half indicated the candidate was female. These files were randomly assigned to the subjects. For each supervisor both the sex associated with the assigned file and the promotion decision were recorded.

1. Is this an observational study or an experiment?
 - Experiment
2. How does the type of study impact what can be inferred from the results?
 - Causation can be tested and confirmed
3. Who or what are the cases?
 - 48 male bank supervisors
4. What is the explanatory variable? What is the response variable? Also say the type of each variable.

Explanatory: Gender of the applicant

Response: Promoted

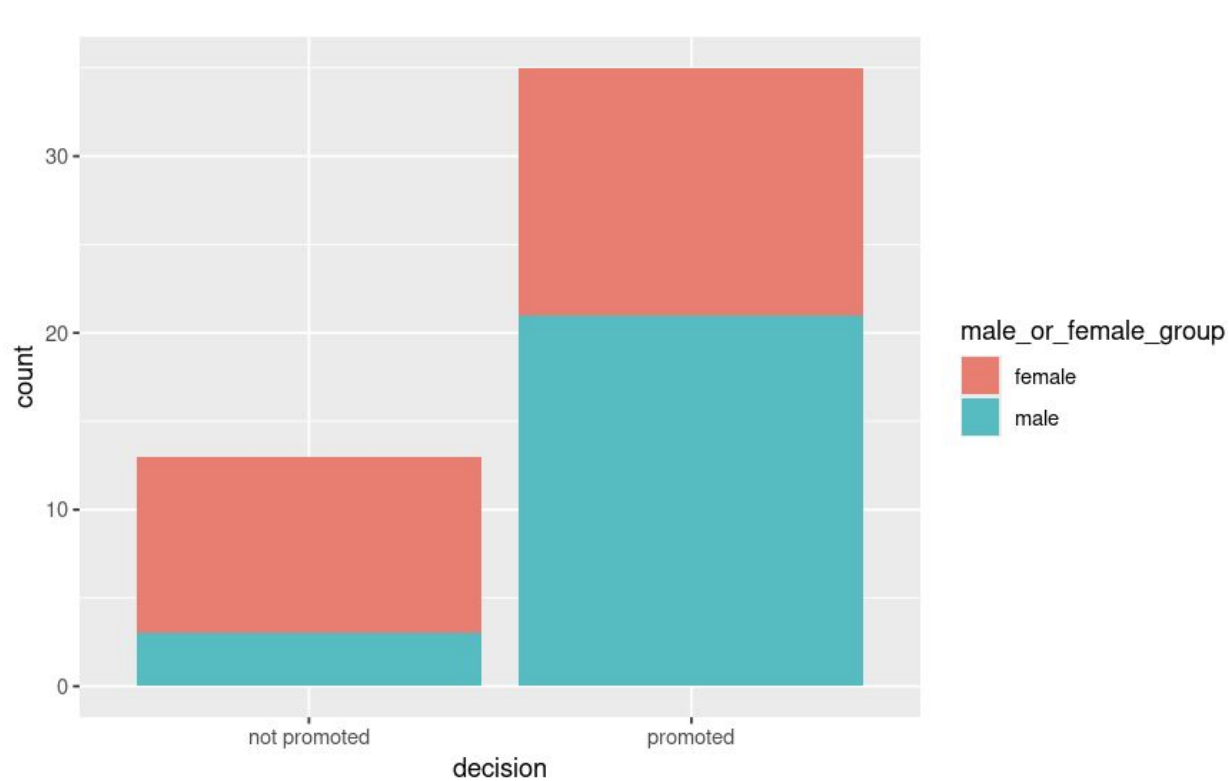
5. In the Posit Cloud file, make a contingency (or two-way) table to summarize the results of the study. Remember to label the columns with the explanatory variable. Put counts (rather than proportions) as the values in your table. Paste your table here.

A tibble: 2 × 3

decision <chr>	female <int>	male <int>
not promoted	10	3
promoted	14	21

2 rows

6. Also make an appropriate graph that summarizes the results of this study. Paste your graph here.



7. What proportion of supervisors made the decision to promote, overall? (*Write your answer as a decimal. Show work.*)

$$(14 + 21)/48 = 0.73$$

8. What proportion of supervisors in the “male file” group decided to promote? (*Write your answer as a decimal. Show work.*)

$$21/24 = 0.88$$

9. What proportion of supervisors in the “female file” group decided to promote? (*Write your answer as a decimal. Show work.*)

$$14/24 = 0.58$$

10. What is the difference in the proportion of supervisors who promoted between each group? Calculate this as “proportion of supervisors in male group who promoted” - “proportion of supervisors in female group who promoted”. (*Write your answer as a decimal. Show work.*)

$$0.88 - 0.58 = 0.30$$

11. What are your initial thoughts on these data? In particular, does it look as if individuals who identify as female are unfairly discriminated against? We will discuss this as a class.

Yes, it does look that individuals who identify as female are unfairly discriminated against.

Group 5

We consider a study investigating sex discrimination in the 1970s, which is set in the context of personnel decisions within a bank. The research question we hope to answer is, “Are females discriminated against in promotion decisions made by their supervisors who are male?” (Rosen and Jerdee 1974)

The participants in this study were 48 male bank supervisors attending a management institute at the University of North Carolina in 1972. They were asked to assume the role of the personnel director of a bank and were each given one personnel file to judge whether the person should be promoted to a branch manager position. The files given to the participants were identical, except that half of them indicated the candidate was male and the other half indicated the candidate was female. These files were randomly assigned to the subjects. For each supervisor both the sex associated with the assigned file and the promotion decision were recorded.

1. Is this an observational study or an experiment?

Experiment

2. How does the type of study impact what can be inferred from the results?

We can determine causality

3. Who or what are the cases?

The male bank supervisors attending a management institute at UNC.

4. What is the explanatory variable? What is the response variable? Also say the type of each variable.

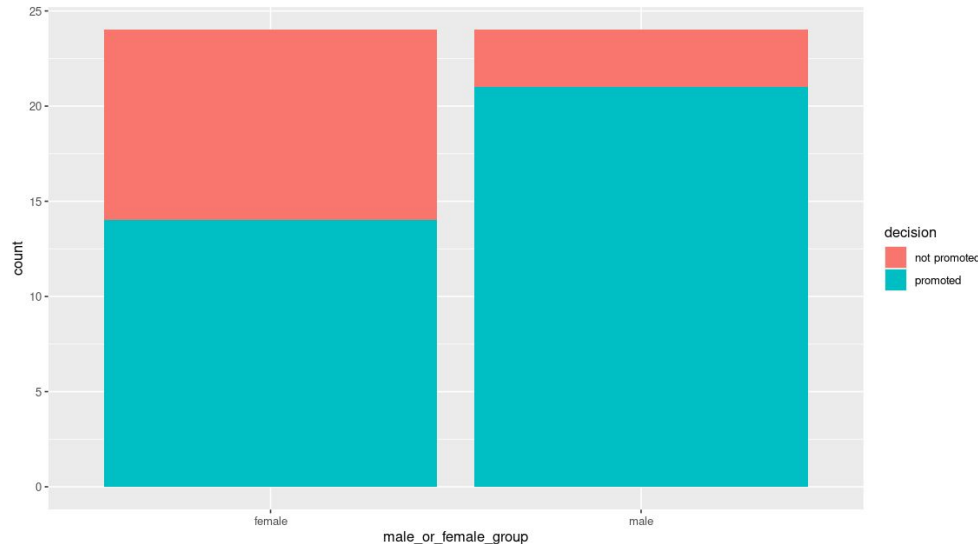
Explanatory: **Gender of the candidate , categorical**

Response: **Whether or not they would get the promotion, categorical**

5. In the Posit Cloud file, make a contingency (or two-way) table to summarize the results of the study. Remember to label the columns with the explanatory variable. Put counts (rather than proportions) as the values in your table. Paste your table here.

decision <chr>	female <int>	male <int>
not promoted	10	3
promoted	14	21

6. Also make an appropriate graph that summarizes the results of this study. Paste your graph here.



7. What proportion of supervisors made the decision to promote, overall? (*Write your answer as a decimal. Show work.*)

$$35/48 = 0.729$$

8. What proportion of supervisors in the “male file” group decided to promote? (*Write your answer as a decimal. Show work.*)

$$21/24 = 0.875$$

9. What proportion of supervisors in the “female file” group decided to promote? (*Write your answer as a decimal. Show work.*)

$$14/24 = 0.583$$

10. What is the difference in the proportion of supervisors who promoted between each group? Calculate this as “proportion of supervisors in male group who promoted” - “proportion of supervisors in female group who promoted”. (*Write your answer as a decimal. Show work.*)

$$(21/24) - (14/24) = 0.2916$$

11. What are your initial thoughts on these data? In particular, does it look as if individuals who identify as female are unfairly discriminated against? We will discuss this as a class.

Yes, since there is approximately a 30% difference in the proportion of females versus males promoted.

Group 6

We consider a study investigating sex discrimination in the 1970s, which is set in the context of personnel decisions within a bank. The research question we hope to answer is, “Are females discriminated against in promotion decisions made by their supervisors who are male?” (Rosen and Jerdee 1974)

The participants in this study were 48 male bank supervisors attending a management institute at the University of North Carolina in 1972. They were asked to assume the role of the personnel director of a bank and were each given one personnel file to judge whether the person should be promoted to a branch manager position. The files given to the participants were identical, except that half of them indicated the candidate was male and the other half indicated the candidate was female. These files were randomly assigned to the subjects. For each supervisor both the sex associated with the assigned file and the promotion decision were recorded.

1. Is this an observational study or an experiment?

This is an experiment

2. How does the type of study impact what can be inferred from the results?

It determines the causality of the results.

3. Who or what are the cases?

The cases are the 48 male bank supervisors attending a management institute

4. What is the explanatory variable? What is the response variable? Also say the type of each variable.

Explanatory:

Response:

5. In the Posit Cloud file, make a contingency (or two-way) table to summarize the results of the study. Remember to label the columns with the explanatory variable. Put counts (rather than proportions) as the values in your table. Paste your table here.

6. Also make an appropriate graph that summarizes the results of this study. Paste your graph here.

7. What proportion of supervisors made the decision to promote, overall? (*Write your answer as a decimal. Show work.*)
8. What proportion of supervisors in the “male file” group decided to promote? (*Write your answer as a decimal. Show work.*)
9. What proportion of supervisors in the “female file” group decided to promote? (*Write your answer as a decimal. Show work.*)
10. What is the difference in the proportion of supervisors who promoted between each group? Calculate this as “proportion of supervisors in male group who promoted” - “proportion of supervisors in female group who promoted”. (*Write your answer as a decimal. Show work.*)
11. What are your initial thoughts on these data? In particular, does it look as if individuals who identify as female are unfairly discriminated against? We will discuss this as a class.

Group 7

We consider a study investigating sex discrimination in the 1970s, which is set in the context of personnel decisions within a bank. The research question we hope to answer is, “Are females discriminated against in promotion decisions made by their supervisors who are male?” (Rosen and Jerdee 1974)

The participants in this study were 48 male bank supervisors attending a management institute at the University of North Carolina in 1972. They were asked to assume the role of the personnel director of a bank and were each given one personnel file to judge whether the person should be promoted to a branch manager position. The files given to the participants were identical, except that half of them indicated the candidate was male and the other half indicated the candidate was female. These files were randomly assigned to the subjects. For each supervisor both the sex associated with the assigned file and the promotion decision were recorded.

1. Is this an observational study or an experiment?
2. How does the type of study impact what can be inferred from the results?
3. Who or what are the cases?
4. What is the explanatory variable? What is the response variable? Also say the type of each variable.

Explanatory:

Response:

5. In the Posit Cloud file, make a contingency (or two-way) table to summarize the results of the study. Remember to label the columns with the explanatory variable. Put counts (rather than proportions) as the values in your table. Paste your table here.

6. Also make an appropriate graph that summarizes the results of this study. Paste your graph here.

7. What proportion of supervisors made the decision to promote, overall? (*Write your answer as a decimal. Show work.*)
8. What proportion of supervisors in the “male file” group decided to promote? (*Write your answer as a decimal. Show work.*)
9. What proportion of supervisors in the “female file” group decided to promote? (*Write your answer as a decimal. Show work.*)
10. What is the difference in the proportion of supervisors who promoted between each group? Calculate this as “proportion of supervisors in male group who promoted” - “proportion of supervisors in female group who promoted”. (*Write your answer as a decimal. Show work.*)
11. What are your initial thoughts on these data? In particular, does it look as if individuals who identify as female are unfairly discriminated against? We will discuss this as a class.

Group 8

We consider a study investigating sex discrimination in the 1970s, which is set in the context of personnel decisions within a bank. The research question we hope to answer is, “Are females discriminated against in promotion decisions made by their supervisors who are male?” (Rosen and Jerdee 1974)

The participants in this study were 48 male bank supervisors attending a management institute at the University of North Carolina in 1972. They were asked to assume the role of the personnel director of a bank and were each given one personnel file to judge whether the person should be promoted to a branch manager position. The files given to the participants were identical, except that half of them indicated the candidate was male and the other half indicated the candidate was female. These files were randomly assigned to the subjects. For each supervisor both the sex associated with the assigned file and the promotion decision were recorded.

1. Is this an observational study or an experiment?
experiment

2. How does the type of study impact what can be inferred from the results?
Determine causation

3. Who or what are the cases?
48 male bank supervisors

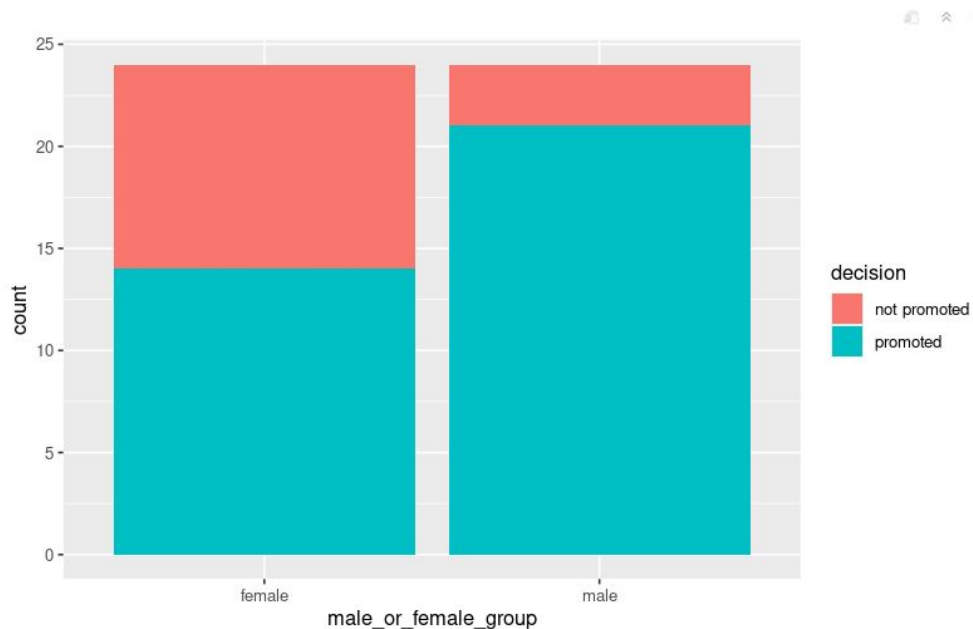
4. What is the explanatory variable? What is the response variable? Also say the type of each variable.
Explanatory: gender on candidate file, categorical
Response: whether the bank manager gives the promotion, categorical

5. In the Posit Cloud file, make a contingency (or two-way) table to summarize the results of the study. Remember to label the columns with the explanatory variable. Put counts (rather than proportions) as the values in your table. Paste your table here.

decision <chr>	female <int>	male <int>
not promoted	10	3
promoted	14	21

2 rows

6. Also make an appropriate graph that summarizes the results of this study. Paste your graph here.



7. What proportion of supervisors made the decision to promote, overall? *(Write your answer as a decimal. Show work.)*

$$35/48 = 0.729166667$$

8. What proportion of supervisors in the “male file” group decided to promote? *(Write your answer as a decimal. Show work.)*

$$21/24=0.875$$

9. What proportion of supervisors in the “female file” group decided to promote? *(Write your answer as a decimal. Show work.)*

$$14/24=0.583333334$$

10. What is the difference in the proportion of supervisors who promoted between each group? Calculate this as “proportion of supervisors in male group who promoted” - “proportion of supervisors in female group who promoted”. *(Write your answer as a decimal. Show work.)*

$$0.875 - 0.583333334 = 0.291666666$$

11. What are your initial thoughts on these data? In particular, does it look as if individuals who identify as female are unfairly discriminated against? We will discuss this as a class.

We can conclude that individuals who identify as females were unfairly discriminated against in this study

Group 9

We consider a study investigating sex discrimination in the 1970s, which is set in the context of personnel decisions within a bank. The research question we hope to answer is, “Are females discriminated against in promotion decisions made by their supervisors who are male?” (Rosen and Jerdee 1974)

The participants in this study were 48 male bank supervisors attending a management institute at the University of North Carolina in 1972. They were asked to assume the role of the personnel director of a bank and were each given one personnel file to judge whether the person should be promoted to a branch manager position. The files given to the participants were identical, except that half of them indicated the candidate was male and the other half indicated the candidate was female. These files were randomly assigned to the subjects. For each supervisor both the sex associated with the assigned file and the promotion decision were recorded.

1. Is this an observational study or an experiment?
Experiment

2. How does the type of study impact what can be inferred from the results?
You can draw conclusions from the data because it's an experiment

3. Who or what are the cases?
candidates

4. What is the explanatory variable? What is the response variable? Also say the type of each variable.

Explanatory: sex of the candidate

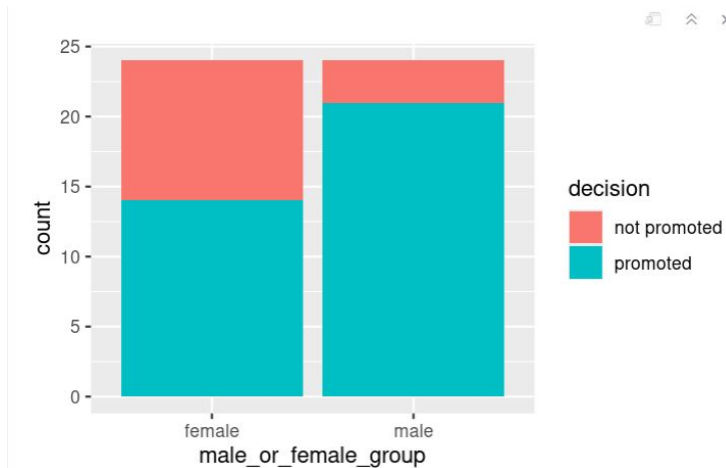
Response: whether or not they are promoted

5. In the Posit Cloud file, make a contingency (or two-way) table to summarize the results of the study. Remember to label the columns with the explanatory variable. Put counts (rather than proportions) as the values in your table. Paste your table here.

decision <chr>	female <int>	male <int>
not promoted	10	3
promoted	14	21

2 rows

6. Also make an appropriate graph that summarizes the results of this study. Paste your graph here.



7. What proportion of supervisors made the decision to promote, overall? *(Write your answer as a decimal. Show work.)*

$$35/48 = .729166666667$$

8. What proportion of supervisors in the “male file” group decided to promote? *(Write your answer as a decimal. Show work.)*

$$21/24 = .875$$

9. What proportion of supervisors in the “female file” group decided to promote? *(Write your answer as a decimal. Show work.)*

$$14/24 = 0.583333333333$$

10. What is the difference in the proportion of supervisors who promoted between each group? Calculate this as “proportion of supervisors in male group who promoted” - “proportion of supervisors in female group who promoted”. *(Write your answer as a decimal. Show work.)*

$$.875 - 0.583333333333 = 0.291666666667$$

11. What are your initial thoughts on these data? In particular, does it look as if individuals who identify as female are unfairly discriminated against? We will discuss this as a class.

Based on the data, it looks likely that there is in fact a bias in terms of female vs male promotion

Group 10

We consider a study investigating sex discrimination in the 1970s, which is set in the context of personnel decisions within a bank. The research question we hope to answer is, “Are females discriminated against in promotion decisions made by their supervisors who are male?” (Rosen and Jerdee 1974)

The participants in this study were 48 male bank supervisors attending a management institute at the University of North Carolina in 1972. They were asked to assume the role of the personnel director of a bank and were each given one personnel file to judge whether the person should be promoted to a branch manager position. The files given to the participants were identical, except that half of them indicated the candidate was male and the other half indicated the candidate was female. These files were randomly assigned to the subjects. For each supervisor both the sex associated with the assigned file and the promotion decision were recorded.

1. Is this an observational study or an experiment?
This study would be an experiment because there is interaction between the participants and the experimenters

2. How does the type of study impact what can be inferred from the results?
This type of study means we can infer that there could be causation in this study

3. Who or what are the cases?
The cases in the study are the male and female candidates

4. What is the explanatory variable? What is the response variable? Also say the type of each variable.

Explanatory: male or female

Response: promoted or not

5. In the Posit Cloud file, make a contingency (or two-way) table to summarize the results of the study. Remember to label the columns with the explanatory variable. Put counts (rather than proportions) as the values in your table. Paste your table here.

6. Also make an appropriate graph that summarizes the results of this study. Paste your graph here.

7. What proportion of supervisors made the decision to promote, overall? (*Write your answer as a decimal. Show work.*)
8. What proportion of supervisors in the “male file” group decided to promote? (*Write your answer as a decimal. Show work.*)
9. What proportion of supervisors in the “female file” group decided to promote? (*Write your answer as a decimal. Show work.*)
10. What is the difference in the proportion of supervisors who promoted between each group? Calculate this as “proportion of supervisors in male group who promoted” - “proportion of supervisors in female group who promoted”. (*Write your answer as a decimal. Show work.*)
11. What are your initial thoughts on these data? In particular, does it look as if individuals who identify as female are unfairly discriminated against? We will discuss this as a class.